

# ARDUINO BASED VEHICLE PARKING COUNTER



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This vehicle counter counts the number of cars and the vacant space available in a parking lot and shows the values on a three-digit dual-colour display. The number of cars is shown in red colour and the space available in green colour.

The circuit design is the same as for the 'Traffic Light Count-Down Timer with Dual-Colour Display' published in May issue of EFY. So if you already have this project, you don't have to buy or assemble a new Arduino board. Just by burning the parking\_counter code given in this article into the ATmega328 microcontroller, the functioning of the board can be changed to a vehicle parking counter. For parts-list, circuit description and AVR programming using Arduino IDE, refer to EFY's May issue.

Fig. 1 shows the block diagram of the Arduino based vehicle parking counter.

## Vehicle loop detector

Almost all parking lots in buildings or malls use loop detectors buried inside the road to detect the movement of vehicles crossing the gates. They detect the vehicles but ignore the gate crossing by humans, bicycles, etc. Usually, these detectors are part of electronic oscillator circuits. The loop detector circuitry and its description are not included here. However, its basic concepts are described here to give you

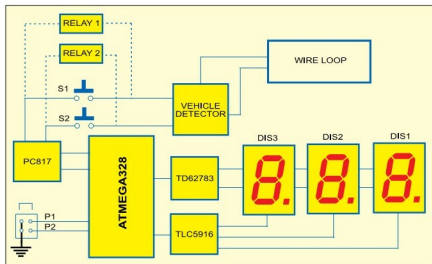


Fig. 1: Block diagram of Arduino based vehicle parking counter

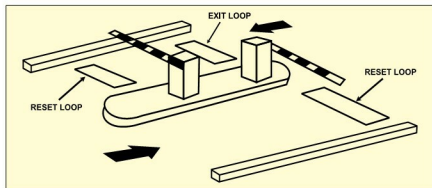


Fig. 2: A typical two-way gate system with boom barriers

some idea about how the loop detector works along with the vehicle parking counter.

A typical two-way gate system with boom barriers is shown in Fig. 2. When you approach a vehicle parking boom gate, you may notice a rectangular scar where the road surface has been cut with a saw and then re-sealed. This is the sensor loop. The loop consists of one or more turns of wire buried in the roadway and connected to an electronic circuit which can detect a

vehicle passing over it.

The boom barrier is raised by pressing a switch to allow vehicles cross the gate. The boom barrier automatically lowers down after the vehicle crosses the gate.

## Vehicle loop detector connections to parking counter

Fig. 3 shows a typical preformed loop with extension cable. The two ends of the loop wire are connected to the loop